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Lucia Faravelli, Editor
Structural Control and Health Monitoring
Wiley

Dear Editor,

Submission of an original research article: "Total-load observability of continuous girder bridges from a single deflection channel: closed-form blind configurations, laboratory verification and sensor placement on an in-service box girder"

We submit the above manuscript for consideration as a Research Article in *Structural Control and Health Monitoring*.

Influence-area bridge weigh-in-motion estimates the gross weight of a vehicle from the area under a single response record. On continuous girders the influence line changes sign over the interior supports, and the positive and negative areas partly cancel. The manuscript treats a question that comes before any weighing algorithm and is, in monitoring terms, a sensor-placement question: for a given continuous girder and sensor position, can one deflection channel observe the total load at all?

The paper makes the following contributions.

1. By reciprocity, the constant that relates the deflection integral to the total load is the deflection at the sensor under a uniformly distributed unit load. The identity is classical; we use it to make the constant available from one static analysis or one static test, before installation. The influence-area idea and its calibration with a weighed vehicle are credited to Ojio and Yamada.

2. Closed forms of this constant for two-span beams with arbitrary span and stiffness ratios and for symmetric three-span beams, which locate the blind configurations exactly. For a three-span beam instrumented at mid side span with uniform rigidity, the blind main-to-side span ratio is 1.700594.

3. Three pre-installation indicators: the normalised constant, the cancellation ratio, and an offset-equivalent load that converts the residual baseline offset of the instrumentation into a load bias for a given gauge.

4. An application to the acceptance-test data of an in-service 55 + 90 + 55 m box girder. Two calibrations reproduce the measured deflections but give side-span gauge indices of 0.18 and 0.48, while the centre-span gauge stays at 0.21–0.23. A residual offset of 0.05 mm biases a 25.06 t truck (the mean of the weighed test trucks) by 35–93 % at the side-span gauge and by 10–11 % at the centre-span gauge. The design model therefore cannot supply the constant of a side-span gauge on such a girder.

The theory is verified against finite element evaluations and on a laboratory two-span steel beam with statically positioned loads, where two- and three-load groups were identified within 1.19 % and 0.70 % of direct weighing. The paper argues why the laboratory scale does not limit these identities and evaluates the scale-dependent terms at the scale of the bridge. The bridge data are static, so the integral identity itself is not validated on a continuous girder, and the manuscript says so.

We believe the paper fits the journal's interest in structural health monitoring theory, sensor technology, signal processing and measurements: it gives a closed-form observability and sensor-placement criterion for a monitoring channel, together with a measurement-error analysis for it. The journal has recently published work on deflection influence lines of continuous beam bridges, which we cite and distinguish.

The manuscript is original, has not been published, and is not under consideration by any other journal. [NEED VALUE: two related papers by the authors are under review at *Structures*; state here whether they share the laboratory rig, data or text with this manuscript and how they are cited, or confirm that there is no overlap.] All authors have approved the submission. Conflicts of interest, funding, data availability and the use of generative AI are declared in the manuscript.

Yours sincerely,

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